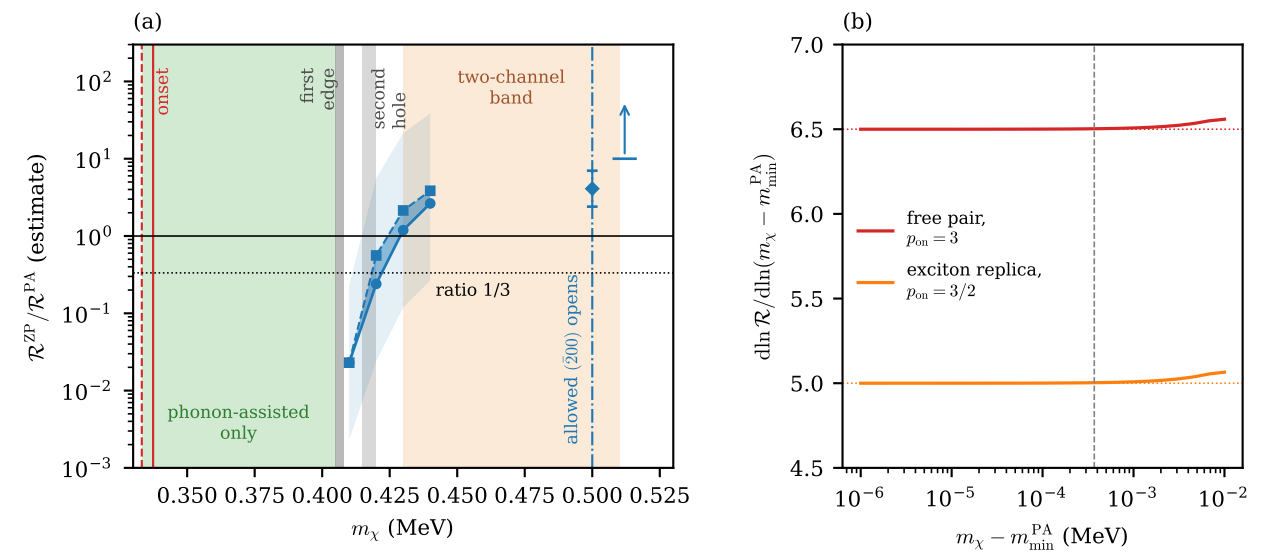
(a)
onset
first edge
second hole
two-channel band
ratio 1/3
allowed (200) opens
phonon-assisted only
$\mathcal{R}^{\rm ZP}/\mathcal{R}^{\rm PA}$ (estimate)
m_χ (MeV)
(b)
free pair, $p_{\rm on} = 3$
exciton replica, $p_{\rm on} = 3/2$
$\mathrm{d}\ln\mathcal{R}/\mathrm{d}\ln(m_\chi - m_{\rm min}^{\rm PA})$
$m_\chi - m_{\rm min}^{\rm PA}$ (MeV)